

1.3 Matrices and Matrix Operations

Rectangular arrays of numbers arise in many contexts beyond augmented matrices. This section studies matrices as objects in their own right, defining addition, subtraction, and multiplication on them.

Matrix Notation and Terminology

Rectangular arrays occur in many contexts. For example, the table below records the hours a student spent on three subjects during a week:

	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	Sun.
Math	2	3	2	4	1	4	2
History	0	3	1	4	3	2	2
Language	4	1	3	1	0	0	2

Suppressing the headings leaves a rectangular array called a **matrix**:

$$\begin{bmatrix} 2 & 3 & 2 & 4 & 1 & 4 & 2 \\ 0 & 3 & 1 & 4 & 3 & 2 & 2 \\ 4 & 1 & 3 & 1 & 0 & 0 & 2 \end{bmatrix}$$

Definition 1

A **matrix** is a rectangular array of numbers. The numbers in the array are called the **entries** of the matrix.

EXAMPLE 1 | Examples of Matrices

$$\begin{bmatrix} 1 & 2 \\ 3 & 0 \\ -1 & 4 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 & 0 & -3 \end{bmatrix}, \quad \begin{bmatrix} e & \pi & -\sqrt{2} \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 4 \end{bmatrix}$$

The **size** of a matrix is given as (rows) $\times$ (columns): the matrices above have sizes 3×2 , 1×4 , 3×3 , 2×1 , and 1×1 . A matrix with one row (like the second above) is a **row vector** (or **row matrix**); one with one column (the fourth above) is a **column vector** (or **column matrix**); the fifth is both. We use capital letters for matrices and lowercase for numerical quantities (**scalars**, always real unless noted otherwise), e.g. $A = \begin{bmatrix} 2 & 1 & 7 \\ 3 & 4 & 2 \end{bmatrix}$.

The entry in row i , column j of A is denoted a_{ij} , so a general $m \times n$ matrix is

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad (1)$$

written compactly as $A = [a_{ij}]_{m \times n}$ or $A = [a_{ij}]$. The entry in row i , column j is also written $(A)_{ij}$, so $(A)_{ij} = a_{ij}$; e.g. for $A = \begin{bmatrix} 2 & -3 \\ 7 & 0 \end{bmatrix}$, $(A)_{11} = 2, (A)_{12} = -3, (A)_{21} = 7, (A)_{22} = 0$.

Row and column vectors are usually denoted by boldface lowercase letters, needing only a single subscript:

$$\mathbf{a} = [a_1 \ a_2 \ \cdots \ a_n] \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

A matrix A with n rows and n columns is a **square matrix of order n** ; the entries $a_{11}, a_{22}, \dots, a_{nn}$ (boxed below) form its **main diagonal**:

$$\begin{bmatrix} \boxed{a_{11}} & a_{12} & \cdots & a_{1n} \\ a_{21} & \boxed{a_{22}} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & \boxed{a_{nn}} \end{bmatrix} \quad (2)$$

Operations on Matrices

So far matrices have only abbreviated linear systems. We now develop an “arithmetic of matrices” – addition, subtraction, and multiplication.

Definition 2

Two matrices are **equal** if they have the same size and their corresponding entries are equal.

EXAMPLE 2 | Equality of Matrices

For $A = \begin{bmatrix} 2 & 1 \\ 3 & x \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \end{bmatrix}$: $A = B$ only if $x = 5$; $A = C$ is impossible for any x , since they differ in size.

Definition 3

If A and B have the same size, the **sum** $A + B$ adds corresponding entries, and the **difference** $A - B$ subtracts them. Matrices of different sizes cannot be added or subtracted.

In symbols, for $A = [a_{ij}]$, $B = [b_{ij}]$ of the same size:

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad \text{and} \quad (A - B)_{ij} = a_{ij} - b_{ij}$$

EXAMPLE 3 | Addition and Subtraction

$$\text{For } A = \begin{bmatrix} 2 & 1 & 0 & 3 \\ -1 & 0 & 2 & 4 \\ 4 & -2 & 7 & 0 \end{bmatrix}, B = \begin{bmatrix} -4 & 3 & 5 & 1 \\ 2 & 2 & 0 & -1 \\ 3 & 2 & -4 & 5 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}:$$

$$A + B = \begin{bmatrix} -2 & 4 & 5 & 4 \\ 1 & 2 & 2 & 3 \\ 7 & 0 & 3 & 5 \end{bmatrix}, \quad A - B = \begin{bmatrix} 6 & -2 & -5 & 2 \\ -3 & -2 & 2 & 5 \\ 1 & -4 & 11 & -5 \end{bmatrix}$$

$A + C, B + C, A - C, B - C$ are all undefined (size mismatch).

Definition 4

For any matrix A and scalar c , the **product** cA multiplies every entry of A by c ; cA is a **scalar multiple** of A .

In symbols, $(cA)_{ij} = c(A)_{ij} = ca_{ij}$.

EXAMPLE 4 | Scalar Multiples

$$\text{For } A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 & 7 \\ -1 & 3 & -5 \end{bmatrix}, C = \begin{bmatrix} 9 & -6 & 3 \\ 3 & 0 & 12 \end{bmatrix}:$$

$$2A = \begin{bmatrix} 4 & 6 & 8 \\ 2 & 6 & 2 \end{bmatrix}, \quad (-1)B = \begin{bmatrix} 0 & -2 & -7 \\ 1 & -3 & 5 \end{bmatrix}, \quad \frac{1}{3}C = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 0 & 4 \end{bmatrix}$$

$(-1)B$ is usually written $-B$.

Multiplying matrices entry by entry (as with addition) turns out not to be useful. Instead:

Definition 5

If A is $m \times r$ and B is $r \times n$, the **product** AB is the $m \times n$ matrix whose (i, j) entry is found by multiplying corresponding entries of row i of A and column j of B , then adding the products.

EXAMPLE 5 | Multiplying Matrices

$$\text{For } A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \ (2 \times 3) \text{ and } B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} \ (3 \times 4), AB \text{ is } 2 \times 4. \text{ E.g., row 2 of } A \text{ with column}$$

3 of B : $(2)(4) + (6)(3) + (0)(5) = 26$; row 1 of A with column 4 of B : $(1)(3) + (2)(1) + (4)(2) = 13$.
Computing all entries:

$$AB = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$$

Matrix multiplication requires the number of columns of the first factor to equal the number of rows of the second; otherwise the product is undefined. Writing the sizes side by side, the inner numbers must

match, and the outer numbers give the product's size:

$$\begin{matrix} A & B & \rightarrow & AB \\ m \times r & r \times n & & m \times n \end{matrix} \quad (3)$$

EXAMPLE 6 | Determining Whether a Product Is Defined

For A (3×4), B (4×7), C (7×3): AB is defined (3×7); BC is defined (4×3); CA is defined (7×4); AC , CB , BA are all undefined.

In general, for $A = [a_{ij}]$ ($m \times r$) and $B = [b_{ij}]$ ($r \times n$), the row-column rule gives

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ir}b_{rj} \quad (5)$$

Partitioned Matrices

A matrix can be **partitioned** into smaller **submatrices** by inserting rules between selected rows/columns. For a general 3×4 matrix A , three useful partitions are into four submatrices, into its three row vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$, or into its four column vectors $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \mathbf{c}_4$.

Matrix Multiplication by Columns and by Rows

Partitioning lets us extract a single row or column of AB without computing the whole product:

$$AB = A[\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_n] = [A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad \cdots \quad A\mathbf{b}_n] \quad (AB \text{ column by column}) \quad (6)$$

$$AB = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_m \end{bmatrix} B = \begin{bmatrix} \mathbf{a}_1 B \\ \mathbf{a}_2 B \\ \vdots \\ \mathbf{a}_m B \end{bmatrix} \quad (AB \text{ row by row}) \quad (7)$$

In words:

$$j\text{th column of } AB = A[j\text{th column of } B] \quad (8)$$

$$i\text{th row of } AB = [i\text{th row of } A]B \quad (9)$$

EXAMPLE 7 | Example 5 Revisited

For A, B as in Example 5, the second column of AB is

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix} = \begin{bmatrix} 27 \\ -4 \end{bmatrix}$$

and the first row of AB is $\begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 12 & 27 & 30 & 13 \end{bmatrix}.$

Matrix Products as Linear Combinations

Definition 6

If $A_1, \dots, A_r$ are matrices of the same size and $c_1, \dots, c_r$ are scalars, then $c_1A_1 + c_2A_2 + \dots + c_rA_r$ is called a **linear combination** of $A_1, \dots, A_r$ with **coefficients** $c_1, \dots, c_r$.

For an $m \times n$ matrix A and $n \times 1$ vector $\mathbf{x} = [x_1 \ \dots \ x_n]^T$,

$$A\mathbf{x} = \begin{bmatrix} a_{11}x_1 + \dots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} \quad (10)$$

which proves:

Theorem 1.3.1

If A is $m \times n$ and $\mathbf{x}$ is $n \times 1$, then $A\mathbf{x}$ is a linear combination of the column vectors of A with coefficients from $\mathbf{x}$.

EXAMPLE 8 | Matrix Products as Linear Combinations

$$\begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix} \iff 2 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

EXAMPLE 9 | Columns of a Product AB as Linear Combinations

From Example 5, $AB = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$. By (6) and Theorem 1.3.1,

the j th column of AB is the linear combination of A 's columns with coefficients from column j of B :

$$\begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 27 \\ -4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 7 \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 30 \\ 26 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 5 \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 13 \\ 12 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

Column-Row Expansion

If $m \times r$ matrix A is partitioned into columns $\mathbf{c}_1, \dots, \mathbf{c}_r$ and $r \times n$ matrix B into rows $\mathbf{r}_1, \dots, \mathbf{r}_r$, then each term $\mathbf{c}_i\mathbf{r}_i$ is $m \times n$, and

$$AB = \mathbf{c}_1\mathbf{r}_1 + \mathbf{c}_2\mathbf{r}_2 + \dots + \mathbf{c}_r\mathbf{r}_r \quad (11)$$

called the **column-row expansion** of AB .

EXAMPLE 10 | Column-Row Expansion

For $AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 \\ -3 & 5 & 1 \end{bmatrix}$: $\mathbf{c}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{c}_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$; $\mathbf{r}_1 = [2 \ 0 \ 4]$, $\mathbf{r}_2 = [-3 \ 5 \ 1]$. So

$$AB = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [2 \ 0 \ 4] + \begin{bmatrix} 3 \\ -1 \end{bmatrix} [-3 \ 5 \ 1] = \begin{bmatrix} 2 & 0 & 4 \\ 4 & 0 & 8 \end{bmatrix} + \begin{bmatrix} -9 & 15 & 3 \\ 3 & -5 & -1 \end{bmatrix} = \begin{bmatrix} -7 & 15 & 7 \\ 7 & -5 & 7 \end{bmatrix}$$

Summarizing Matrix Multiplication

Five equivalent ways to compute a matrix product:

1. Entry by entry (Definition 5)
2. Row-column method (Formula (5))
3. Column by column (Formula (6))
4. Row by row (Formula (7))
5. Column-row expansion (Formula (11))

Matrix Form of a Linear System

For m equations in n unknowns, equal matrices means the system is equivalent to a single matrix equation:

$$\begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \iff \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Naming these matrices A , $\mathbf{x}$, $\mathbf{b}$, the system becomes the single equation

$$A\mathbf{x} = \mathbf{b}$$

A is the **coefficient matrix**; adjoining $\mathbf{b}$ as a last column gives the augmented matrix $[A \mid \mathbf{b}]$.

Transpose of a Matrix

We close with two operations that have no analog in ordinary arithmetic.

Definition 7

For any $m \times n$ matrix A , the **transpose** A^T is the $n \times m$ matrix obtained by interchanging the rows and columns of A : the first column of A^T is the first row of A , and so on.

EXAMPLE 11 | Some Transposes

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ 5 & 6 \end{bmatrix}, \quad C = [1 \ 3 \ 5], \quad D = [4]$$

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \\ a_{14} & a_{24} & a_{34} \end{bmatrix}, \quad B^T = \begin{bmatrix} 2 & 1 & 5 \\ 3 & 4 & 6 \end{bmatrix}, \quad C^T = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \quad D^T = [4]$$

The entry in row i , column j of A^T equals the entry in row j , column i of A :

$$(A^T)_{ij} = (A)_{ji} \quad (14)$$

For a square matrix, A^T can also be obtained by “reflecting” A about its main diagonal – interchanging entries symmetric about that diagonal:

$$A = \begin{bmatrix} 1 & -2 & 4 \\ 3 & 7 & 0 \\ -5 & 8 & 6 \end{bmatrix} \longrightarrow A^T = \begin{bmatrix} 1 & 3 & -5 \\ -2 & 7 & 8 \\ 4 & 0 & 6 \end{bmatrix} \quad (15)$$

Trace of a Matrix

Definition 8

For a square matrix A , the **trace** of A , written $\text{tr}(A)$, is the sum of the entries on its main diagonal. The trace is undefined if A is not square.

EXAMPLE 12 | Trace

$$\text{For } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 2 & 7 & 0 \\ 3 & 5 & -8 & 4 \\ 1 & 2 & 7 & -3 \\ 4 & -2 & 1 & 0 \end{bmatrix} :$$

$$\text{tr}(A) = a_{11} + a_{22} + a_{33}, \quad \text{tr}(B) = -1 + 5 + 7 + 0 = 11$$

Exercise Set 1.3

1. Suppose A, B, C, D, E have sizes $(4 \times 5), (4 \times 5), (5 \times 2), (4 \times 2), (5 \times 4)$. Determine whether each expression is defined, and if so, give the size of the result.

(a) BA (b) AB^T (c) $AC + D$ (d) $E(AC)$ (e) $A - 3E^T$ (f) $E(5B + A)$

3. Using $A = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$, $E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$,

compute if defined:

(a) $D + E$ (b) $D - E$ (c) $5A$ (d) $-7C$ (e) $2B - C$ (f) $4E - 2D$

(g) $-3(D + 2E)$ (h) $A - A$ (i) $\text{tr}(D)$ (j) $\text{tr}(D - 3E)$ (k) $4\text{tr}(7B)$ (l) $\text{tr}(A)$

4. Using the matrices from Exercise 3, compute if defined:

(a) $2A^T + C$ (b) $D^T - E^T$ (c) $(D - E)^T$ (d) $B^T + 5C^T$ (e) $\frac{1}{2}C^T - \frac{1}{4}A$ (f) $B - B^T$

(g) $2E^T - 3D^T$ (h) $(2E^T - 3D^T)^T$ (i) $(CD)E$ (j) $C(BA)$ (k) $\text{tr}(DE^T)$ (l) $\text{tr}(BC)$

7. Using $A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$, use the row or column method to find:
- (a) the first row of AB (b) the third row of AB (c) the second column of AB
 (d) the first column of BA (e) the third row of AA (f) the third column of AA
9. Using A, B from Exercise 7:
- (a) Express each column of AA as a linear combination of the columns of A .
 (b) Express each column of BB as a linear combination of the columns of B .
11. Find matrices $A, \mathbf{x}, \mathbf{b}$ expressing each system as $A\mathbf{x} = \mathbf{b}$, and write out the equation.
- (a) $2x_1 - 3x_2 + 5x_3 = 7$, $9x_1 - x_2 + x_3 = -1$, $x_1 + 5x_2 + 4x_3 = 0$
 (b) $4x_1 - 3x_3 + x_4 = 1$, $5x_1 + x_2 - 8x_4 = 3$, $2x_1 - 5x_2 + 9x_3 - x_4 = 0$, $3x_2 - x_3 + 7x_4 = 2$
13. Express each matrix equation as a system of linear equations.
- (a) $\begin{bmatrix} 5 & 6 & -7 \\ -1 & -2 & 3 \\ 0 & 4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 5 & -3 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -9 \end{bmatrix}$
15. Find all values of k , if any, satisfying

$$\begin{bmatrix} k & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = 0$$

17. Use the column-row expansion of AB to express the product as a sum of matrix products, for $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & 2 \\ -2 & 3 & 1 \end{bmatrix}$.
21. For the linear system in Example 5 of Section 1.2, express the general solution obtained there as a linear combination of column vectors containing only numerical entries. [*Suggestion:* write the general solution as one column vector, split it into column vectors each containing at most one parameter, then factor out the parameters.]
23. Solve for a, b, c, d :
- $$\begin{bmatrix} a & 3 \\ -1 & a+b \end{bmatrix} = \begin{bmatrix} 4 & d-2c \\ d+2c & -2 \end{bmatrix}$$
25. (a) Show that if A has a row of zeros and B is any matrix for which AB is defined, then AB also has a row of zeros.
 (b) State and justify a similar result for a column of zeros.
27. How many 3×3 matrices A satisfy $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+y \\ x-y \\ 0 \end{bmatrix}$ for all x, y, z ?
29. A matrix B is a **square root** of A if $BB = A$.
- (a) Find two square roots of $A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$.
 (b) How many different square roots can you find of $A = \begin{bmatrix} 5 & 0 \\ 0 & 9 \end{bmatrix}$?
 (c) Do you think every 2×2 matrix has at least one square root? Explain.

30. Let 0 denote the 2×2 zero matrix.

(a) Is there a 2×2 matrix $A \neq 0$ with $AA = 0$? Justify your answer.

(b) Is there a 2×2 matrix $A \neq 0$ with $AA = A$? Justify your answer.

32. Find a 4×4 matrix $A = [a_{ij}]$ satisfying:

$$(a) a_{ij} = i + j \quad (b) a_{ij} = i^{j-1} \quad (c) a_{ij} = \begin{cases} 1 & |i - j| > 1 \\ -1 & |i - j| \leq 1 \end{cases}$$

33. Type I, II, III items cost \$1, \$2, \$3 each; the table gives the numbers purchased each month:

	Type I	Type II	Type III
Jan.	3	4	3
Feb.	5	6	0
Mar.	2	9	4
Apr.	1	1	7

What does the following product represent?

$$\begin{bmatrix} 3 & 4 & 3 \\ 5 & 6 & 0 \\ 2 & 9 & 4 \\ 1 & 1 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

35. **(Working with Proofs)** Prove: if A, B are $n \times n$, then $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$.